The trip may be long; as much as 10 billion light-years' distance. But when you move at near light speed, your time flows far more slowly than on Earth. If you are close enough to light speed, you can make it to your target in a few years or less, as measured by you—slowing down with the aid of a highly elliptical black-hole binary at your target, if you can find one! See Figure 13.6.

You can return home by the same method. But your homecoming may not be pleasant. Billions of years will have passed at home, while you have aged only a few years. Imagine what you find.

These types of slingshots could provide a means for spreading a civilization across the great reaches of intergalactic space. The principal obstacle (perhaps insurmountable!) is finding, or making, the needed black hole binaries. The launch binary might not be a problem if you are a sufficiently advanced civilization, but the slow-down binary is another matter.

What happens to you if there is no slow-down binary, or there is one, but your aim is bad and you miss it? This is a tricky question because of the expansion of the universe. Think about it.

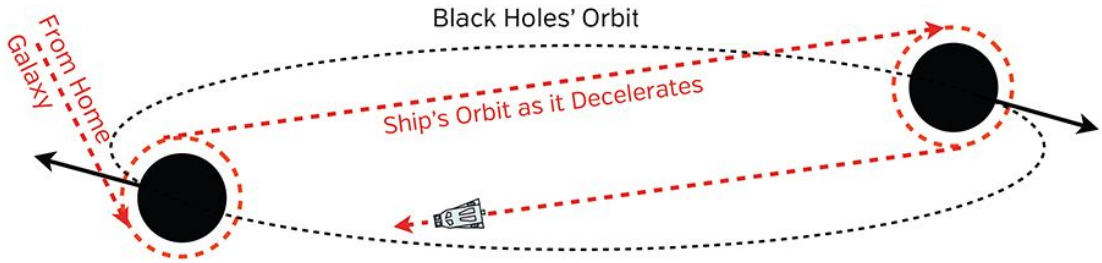


Fig. 13.6. Slowing down by slingshots in a target black-hole binary.

As exciting as these three far-future propulsion systems may seem, they truly *are* far future. Using twenty-first-century technology, we are stuck with thousands of years to reach other solar systems. The only